An energy budget

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Connect work, motion, heating and an honest model conclusion.

SUCCESS CRITERIA - I can...

1. Keep all terms in consistent energy units.
2. Distinguish mechanical-energy change from total-energy loss.
3. Report assumptions with the energy budget.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A 0.3 kg object moving at 5 m/s stops through friction over 0.7 m. Find energy transformed and average opposing-force magnitude.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $E_i + E_{in} = E_f + E_{out}$.
i/f = initial/final stored energy; in/out = transfers across the chosen system boundary. Account for every form.

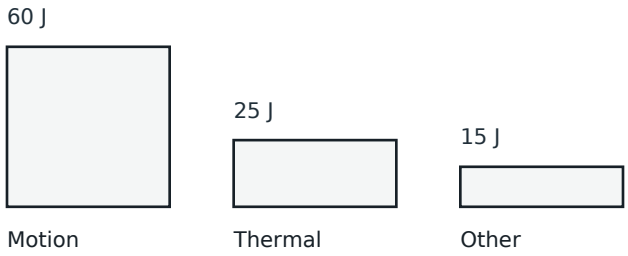
Worked reasoning: An input of 100 J gives 60 J useful motion, 25 J thermal transfer and 15 J sound/other transfers. The account balances because 60+25+15=100 J.

Your turn: Draw proportional energy arrows and calculate the useful fraction.

Marzano check - circle: ① ② ③

03 Represent and interpret

A modeled system starts with 100 J. It ends with 60 J motion energy, 25 J thermal energy and 15 J sound/deformation. Draw an energy account and check the total.



Bar height represents energy. Read the value and unit above each bar.

Marzano check - circle: ① ② ③

04 Solve and justify

A 1000 kg toy probe has$K=8\times10^6$ J and gravitational $U=-6\times10^6$ J. Find total mechanical energy. If it loses 3×10^6 J to drag, what is the total afterward? Discuss bound/escape energy in this ideal reference.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Analyze the reasoning

What information is missing if an impact report gives only total kinetic energy and says all of it heats the ground?

Marzano check - circle: 1 2 3

06 Choose a method

With GMm=192 J·m, move an object from r=4 to r=8 m. Find work by gravity using U=-GMm/r.

Marzano check - circle: 1 2 3

07 Exit check

A 2 kg cart descends 3 m from rest with g=10. Losses are 12 J. Find bottom speed.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A device takes 600 W for 5 s and raises a 10 kg load 12 m usingg=10. Find useful work, efficiency and remaining energy.

Marzano check - circle: 1 2 3